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The cost and performance advantage that you get out of ASICs definitely tilts the balance towards ASICs. FPGAs come into play when volumes are small and time-to-market is critical. So for high-volume or cost-sensitive applications, ASIC is probably the way to go

— Milind Gandhe, Associate Vice President, Semiconductor Business Line, SASKEN

chip are said to be increasing, especially since technology nowadays is ever more complex. As we move towards advanced nodes, cost considerations multiply exponentially. For instance, the development cost for ASIC at 28nm is 40 per cent more than at 40nm. At 20nm, it is estimated to be 70 per cent over that of 28nm.

"As a result of the prohibitive costs in building custom chips, the ASIC and ASSP industry is seeing much less design starts than a few years ago. This is forcing them to narrow down to ultra-high-volume applications like mobile phones, PCs and video games where the investments can be justified. Other areas where custom chips are still prevalent and needed are where you need significant mixed-signal capabilities. Today, FPGAs are able to serve the vast majority of the rest of the high-volume market segments and applications. Xilinx is routinely shipping millions of its low-cost FPGA units to customers in high-volume consumer applications like flat-panel and 3D TVs," adds Varma.

Best of both the worlds

While FPGAs are excellent for designing and prototyping digital logic into medium-volume, medium-density applications, their high unit cost makes things difficult. On the other hand, the low unit cost of ASICs is one of the main reasons why these are considered for high-volume manufacturing.

What if there were a way to get the best of both the worlds? Well, designing a new product around FPGA allows design modifications to be quickly made throughout the development process. Once this de-

sign is complete and approved for production, the FPGA design can be migrated to an ASIC design and then produced, cutting the production unit cost greatly.

"The low non-recurring engineering charges associated with a mid-range ASIC solution coupled with a much lower unit cost make this strategy a powerful tool in achieving low overall costs, giving users a competitive cost advantage in the market," explains Terry Danzer, digital ASIC product marketing manager, Applications Product Group, ON Semiconductor.

To optimise the migration process and ease it forward, Danzer suggest several things that can be considered during the initial design flow: "Careful selection of IP early during the design phase is essential. In addition, developing the FPGA and ASIC in a parallel design flow will help to speed up the process. Finally, planning for portability to an ASIC from the beginning of the project will help to speed up time-to-market and decrease costs. Good design practices such as the use of synchronous design techniques will enable the design to be ported across many different technology platforms. Finally, one of the most important things a design team can do is have good documentation of the design. With a little time and effort spent in the early stages of the design, the migration will require minimal engineering resources. The migration can result in a drop in replacement parts. For additional cost reduction, the design can be ported to less expensive technology nodes and/or packaging." ●

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